

Nishant Sharma

Email: hellonishantsh@gmail.com
New York, NY

Portfolio: hellonish.dev
LinkedIn: linkedin.com/in/nishantsh20/
Github: github.com/hellonish

Projects

Singularity - AI Deep Research Platform | Mar 2026 – Apr 2026 | singularity.hellonish.dev

- Production multi-agent LLM pipeline. Planning runs before retrieval. Deployed on AWS at \$26/month. Zero inference cost via BYOK architecture.
- Designed Phase B to A pipeline: planner generates full report tree before retrieval begins - every search query targets a planned section, not a vague topic. Measurably reduced hallucination and unfocused fanout.
- Built JWT family rotation with reuse detection compromised refresh token revokes entire family, forces re-auth. SSE auth uses separate 30-second single-use JWT to avoid long-lived tokens in query params.
- Replaced sentence-transformers with fastembed/ONNX 1.4 GB lighter, no GPU dependency, production-viable on a 2 GB VM.
- Redis sliding-window rate limiter (ZREMBYRANGE + ZCARD + ZADD) in one atomic pipeline transaction. Fails open on Redis outage.
- Orphan job recovery on worker restart - scans for crashed jobs, marks failed, publishes SSE error. 6-container Docker stack on AWS t3-small.

Verifier-Guided Reasoning with Small Language Models | Sep 2025 – Dec 2025 | [paper](#)

- Developed a generator-verifier system to enhance mathematical reasoning, fine-tuning Phi-2 (2.7B) as a generator on GSM8K and training TinyLLaMA (1.1B) as a step-level verifier.
- Utilized QLoRA and Chain-of-Thought supervision to detect reasoning errors without modifying generator weights, achieving 64.1% Pass@1 accuracy with <0.3% trainable parameters.
- Demonstrated that optimized system design and step-level verification on the PRM800K dataset outperform larger models in reliable reasoning tasks under tight compute constraints.

Snap2Caption - ML Systems for Caption Generation | Mar 2025 – May 2025 | [github](#)

- Built an end-to-end image captioning service that generates Instagram-ready captions and hashtags via a FastAPI inference API with <2s P90 latency.
- Fine-tuned LLaVA-1.5/1.6 (7B) using LoRA on 100k urban images, enabling efficient multimodal training with reproducible, containerized workflows.
- Deployed and operated the system on GPU infrastructure, automating provisioning with Terraform and implementing MLflow-based experiment tracking and monitoring.

Work Experience

Machine Learning Teaching Assistant | NYU Tandon School of Engineering | NY, USA | Sep 2025 - Present

- Standardized PyTorch training pipelines (data loading, training, evaluation) used by 100+ students across course offerings.
- Led debugging sessions on CNN optimization, backpropagation, autograd helping students diagnose training failures and performance bottlenecks.

Co-Founder | Macverin Technologies | www.macverin.com | Jul 2022 - Aug 2024

- Architected distributed microservices (FastAPI + Node.js) serving 1K+ DAU for 10+ clients; maintained 99.9% system availability.
- Built full-stack products end-to-end: TypeScript/React/Next.js frontends, ML inference backends, AWS deployments, automated CI/CD pipelines.

Education

- **New York University - Tandon School of Engineering** | New York, US | Sep 2024 - Expected May 2026
Master of Science in Computer Engineering · GPA: 3.67
FOCUS: LANGUAGE MODELS, MULTIMODAL MODELS, TRAINING & FINETUNING
- **Dr. A.P.J. Abdul Kalam Technical University** | Delhi, India | Aug 2020 - Jun 2024
B.Tech Computer Science and Engineering · 1st Division with Distinction

Technical Skills

- **AI/ML:** RAG pipelines, LLM finetuning (LoRA/QLoRA), RLHF (GRPO/PPO/DPO), Supervised finetuning, PyTorch, ONNX, TensorRT, **DSPy**, LangGraph, Vector stores (Qdrant, Chroma, Pinecone), MLflow
- **Backend:** FastAPI, PostgreSQL, Redis, ARQ, asyncpg, SQLAlchemy, Node.js, Django, REST, gRPC, WebSockets
- **Infra:** Docker, Kubernetes, AWS, GCP, Terraform, GitHub Actions, Prometheus, Grafana, TorchServe
- **Frontend:** Next.js, React, TypeScript Languages: Python, TypeScript, Java, C++, SQL